

# Analysis of Variance & Pairwise Comparisons

CSU, Chico Math 351

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# outline

Recap

Decision Making is hard

Multiple Comparisons – dangerous implications

Multiple Comparison Approaches

- Bonferroni

- Tukey Method

Examples

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## recap, ANOVA

ANOVA compares the means of  $k$  groups in the hypothesis

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

$H_A$  : at least one mean is different

The test brought us a new statistic with a new standard error.

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# Hypothesis Testing, the decision

We conclude a formal hypothesis test by making a decision between  $H_0$  and  $H_1$ . But did we decide correctly?

# Hypothesis Testing, decision errors

We essentially made a choice, but our choice could be correct or incorrect.

|            | fail to reject $H_0$ | reject $H_0$ |
|------------|----------------------|--------------|
| $H_0$ true | Correct              | Type I Error |
| $H_A$ true | Type II Error        | Correct      |

# Hypothesis Testing, errors

## Type I Error

A Type I (1) Error is rejecting the null hypothesis when  $H_0$  is actually true.

## Type II Error

A Type II (2) Error is failing to reject the null hypothesis when the alternative is actually true.



# Hypothesis Testing, error trade offs

Consider the following example: how could we reduce the Type II Error rate in US courts? What influence would this have on the Type I Error rate?

- ▶ Lower Type II Error by convicting more (guilty) people.
- ▶ What then of wrongful convictions, i.e. Type I Error rates?

A trade off exists when considering both types of error.

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# Multiple Comparisons

Running multiple tests on pairs of variables is called **multiple comparisons** and increases the Type I Error rate. Suppose you have data across a categorical variable with  $k = 5$  levels, and you want to compare the means across groups. How many pairwise comparisons exist, and what is the probability that at least one of them gives us a Type I error?

There would be 10 pairwise comparisons—the number of ways to pick 2 levels out of 5 possible ones, or  $\binom{5}{2}$ . Thus, if we let  $\alpha = 0.05$ , the chance of incorrectly rejecting at least one true  $H_0$  would be:

$$\begin{aligned} &1 - P(\text{no Type I errors}) \\ &= 1 - (.95)^{10} \\ &= 0.401 \end{aligned}$$

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# Bonferroni Correction

Our goal is to test the quality of all pairs of individual means:

$H_0 : \mu_i = \mu_j$  vs.  $H_1 : \mu_i \neq \mu_j$  for all  $i \neq j$ .

One way of doing so, the **Bonferroni correction**, uses a smaller significance, calculated from the significance level you originally chose.

## Bonferroni correction

Given a pre-chosen significance level  $\alpha$ , the Bonferroni correction specifies a new significance level to use

$$\alpha^* = \alpha/K,$$

where  $K = \binom{k}{2}$  and  $k$  is the number of groups.

While this is easy to implement, it is very conservative compared to other methods for pairwise comparisons.

## Tukey Method - Premise

The Tukey Method does this using confidence intervals for  $\mu_i - \mu_j$  and the ratio  $R/S$ , where  $R$  is the range of a set of normally distributed random variables, and  $S$  is an estimator for their true standard deviation.

Let  $W_1, W_2, \dots, W_k$  be a set of  $k$  independent, normally distributed random variables with mean  $\mu$  and variance  $\sigma^2$ , with  $R$  denoting their range:

$$R = \max_i W_i - \min_i W_i$$

Let  $S^2$  be a chi-square random variable with  $\nu$  degrees of freedom, independent of the  $W_i$ s, where  $E(S^2) = \sigma^2$ . The **studentized range**  $Q_{k,\nu}$  is the ratio:

$$Q_{k,\nu} = R/S$$

## Tukey Method - using ptukey()

Let  $\bar{Y}_j, j = 1, 2, \dots, k$  be the  $k$  sample means in a completely randomized one-factor design. Let  $n$  be the total sample size, and let  $\mu_j$  be the true means for all  $j = 1, 2, \dots, k$ . The probability is  $1 - \alpha$  that all  $\binom{k}{2}$  differences  $\mu_i - \mu_j$  will simultaneously satisfy the inequalities:

$$\bar{Y}_{.i} - \bar{Y}_{.j} - D\sqrt{MSE} < \mu_i - \mu_j < \bar{Y}_{.i} - \bar{Y}_{.j} + D\sqrt{MSE}$$

where

$$D = Q_{1-\alpha, k, n-k} / \sqrt{r}$$

where  $r$  is the sample size for each group. Note that these are assumed to be equal—for unequal cases, a different approach is needed that requires estimating standard deviation for each comparison (not discussed here).

The studentized range  $Q_{1-\alpha, k, n-k}$  can be calculated in RStudio by using: **qtukey(1- $\alpha$ , k, n-k)**.

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## Example 1 from Larsen & Marx

A certain fraction of antibiotics injected into the bloodstream are bound to serum proteins. This phenomenon bears directly on the effectiveness of the medication, because the binding decreases the systemic uptake of the drug. The table below lists the binding percentages in bovine serum measures for five widely prescribed antibiotics. Which have similar binding properties, and which are different?

|                | Penicillin | Tetracyc. | Streptomycin | Erythrom. | Chloramphen. |
|----------------|------------|-----------|--------------|-----------|--------------|
|                | 29.6       | 27.3      | 5.8          | 21.6      | 29.2         |
|                | 24.3       | 32.6      | 6.2          | 17.4      | 32.8         |
|                | 28.5       | 30.8      | 11.0         | 18.3      | 25.0         |
|                | 32.0       | 34.8      | 8.3          | 19.0      | 24.2         |
| $T_j$          | 114.4      | 125.5     | 31.3         | 76.3      | 111.2        |
| $\bar{Y}_{ij}$ | 28.6       | 31.4      | 7.8          | 19.1      | 27.8         |

## Example 1 from Larsen & Marx

From our data,  $k = 5$ ,  $r = 4$ , and  $n = k*r = 20$ .

First we can find that SSE is:

$$SSE = \sum_{j=1}^5 \sum_{i=1}^4 (Y_{ij} - \bar{Y}_{.j})^2 = 135.83$$

$$\text{And } MSE = \frac{SSE}{(n-k)} = \frac{135.83}{20-5} = 9.06$$

The studentized range would be  $Q_{1-\alpha, k, n-k} = Q_{0.95, 5, 15} = \text{qtukey}(0.95, 5, 15) = 4.3670$ .

Thus we have  $D = 4.367/\text{sqrt}(4) = 2.1835$  and  $D\sqrt{MSE} = 6.5704$ .

## Example 1 from Larsen & Marx

This gives us everything we need to construct confidence intervals for the differences. For instance, take  $\mu_1 - \mu_2$ :

$$\bar{Y}_{.1} - \bar{Y}_{.2} - (D\sqrt{MSE}) < \mu_1 - \mu_2 < \bar{Y}_{.1} - \bar{Y}_{.2} + (D\sqrt{MSE})$$

$$28.6 - 31.4 - (6.5704) < \mu_1 - \mu_2 < 28.6 - 31.4 + (6.5704) \\ (-9.37 < \mu_1 - \mu_2 < 3.77)$$

Since our resulting Tukey interval contains 0, we fail to reject  $H_0$  for this pair—there is no significant difference.

For  $\mu_1 - \mu_3$ :

$$28.6 - 7.8 - (6.5704) < \mu_1 - \mu_2 < 28.6 - 7.8 + (6.5704) \\ (14.23 < \mu_1 - \mu_2 < 27.38)$$

This interval does **not** contain 0—the two group means are significantly different.

## Exercise 1 from Larsen & Marx

Using the smoking heart rate data from the ANOVA part 1 notes, determine whether there is a significant pairwise difference between moderate smokers and heavy smokers (at  $\alpha = 0.05$ ):

- ▶ Using the Bonferroni correction
- ▶ Using the Tukey method

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## references I

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Adapted from notes by Dr. Gray & Dr. Roualdes.